Maxim Beauchemin

+1 (437) 770-2227 | maxim.beauchemin07@gmail.com | LinkedIn | Portfolio | Github

Technical Skills

Programming: C, C++, C#, Java, Python, Bash, Excel VBA
Firmware: FreeRTOS, STM32, CAN, UART, I2C, GoogleTest, Fake Function Framework
Automation: Studio 5000, FactoryTalk, OPC UA, Weintek HMI
Tools & Platforms: Git/GitHub, Docker, WSL, SolidWorks, AutoCAD
Languages: English and French, fluent

Education

University of Waterloo Waterloo, ON
*BASc, Mechatronics Engineering — **Cum. Average: 98%*** *Sep 2025 – Present*

- **Dean's Scholar Award Winner**: highest academic achiever in program
- Relevant Courses: Digital Computation (97%), Data Structures and Algorithms (94%), Circuits (100%)

Experience

Controls Programming Student Barrie, ON
Innovative Automation Inc. *Sep 2026 – Dec 2026*

- Headed HMI development for a production line of six cells, including safety and operator main screens, and supporting features with PLC programming.
- Developed an Excel/VBA tool to automatically generate I/O trackers, enabling real-time value and description updates in PLCs over an RTD server, saving **one work day** per project.
- Improved part validation using computer vision at six stations, achieving **99%** accuracy during the testing.

Software Developer Student Barrie, ON
Innovative Automation Inc. *Jan 2026 – Apr 2026*

- Customized Epicor ERP workflow in C#, reducing manual expense entry by **50%**.
- Created a SolidWorks add-in to fix imported sketches from external file formats, saving an hour of work per week.

Coding Instructor Toronto, ON
Codezilla Kids Ltd. *Feb 2025 – Aug 2025*

- Collaborated with teachers to give parent presentations and hold a 5-star Google rating for the Leaside location.

Projects

Waterloo Rocketry Design Team Waterloo, ON
Controls Subteam *Jun 2026 – Present*

- Led redesign of error log system, encoding entries from **128 to 32 bytes**, and aggregated repeated logs to avoid flooding from 400 Hz tasks.
- Designed FreeRTOS motor telemetry in C, sending current, angle, and temperature over CAN at frequencies set by flight state.
- Wrote C++ unit tests in Docker for canard fin actuation on STM32H733 firmware with GoogleTest and Fake Function Framework (FFF) to simulate successful and unsuccessful UART, CAN, and I2C transmissions.

Autonomous Tic-Tac-Toe Robot Waterloo, ON
C++, SolidWorks, 3D Printing, Sensors *Nov 2025*

- Integrated colour sensing to detect board state and motor control to autonomously place pieces determined by game logic, obtaining **0%** loss rate across 50 games and consistent accurate placement.
- Modeled and 3D printed components in SolidWorks: game pieces, game board, and motor mount.

Extracurriculars

- **2nd degree black belt** and volunteer instructor in Shotokan karate
- Waterloo Poker Club and Chess Club (1800 rating)